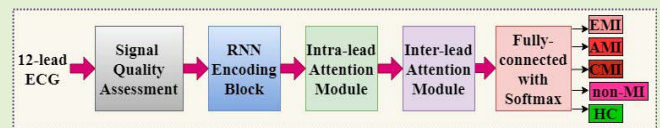


Myocardial Infarction Severity Stages Classification From ECG Signals Using Attentional Recurrent Neural Network

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Abstract—Myocardial infarction (MI) is a lethal heart condition that occurs due to the lack of blood flow to the heart tissues. Based on the time from symptoms onset, it is categorized into three severity stages: early MI (EMI), acute MI (AMI), and chronic MI (CMI). Electrocardiogram (ECG) signals are often used to diagnose MI with pathological changes in its characteristics. In clinical practice, accurate diagnosis and risk-stratification are essential to optimize various treatment strategies, hence clinical outcome. However, most automated methods focus only on identifying MI patients from healthy controls (HC). Therefore, in this paper, we propose a novel multi-lead diagnostic attention-based recurrent neural network (MLDA-RNN) for automated diagnosis of the three MI severity stages from HC subjects. The method systematically processes the 12-lead ECGs to capture the multi-scale temporal dependencies from each ECG leads for improved classification. Specifically, we first employ the RNNs to encode the temporal variations in the 12-lead ECG signals. These encoded vectors are fed to the intra-lead attention module to summarize the within-lead discriminative vectors to obtain lead-attentive representations. Then, the inter-lead attention module aggregates these representative vectors based on their clinical relevance to obtain a high-level feature representation for a reliable diagnosis. Using 12-lead ECGs from the PTBDB and STAFF III datasets, we achieved an overall accuracy of 97.79% without compromising on the class-wise detection rates. With improved performance, the MLDA-RNN also shows promising results for model interpretability as the learned attention weights often correlate with clinicians' way of diagnosing MI severity stages.



Index Terms—Electrocardiogram (ECG), myocardial infarction, recurrent neural networks, attention mechanism.

I. INTRODUCTION

MYOCARDIAL infarction (MI) is one of the leading causes of death among cardiovascular diseases (CVDs) globally [1]. According to the World Health Organization (WHO), MI claims nearly 9 million deaths every year, and this number is projected to be more than 12 million by 2030 [1]. It occurs due to prolonged blockage of coronary arteries by progressive atherosclerotic plaques, thereby reducing oxygenated blood and nutrients to the myocardium [2]. The lack of oxygen causes damage to the myocardial tissues, which slowly expands across the “myocardium at risk” and finally manifests as complete necrosis [3] (see Fig. 1(a)). This progression of MI is broadly categorized into three severity stages

i.e., early MI (EMI), acute MI (AMI), and chronic MI (CMI) based on the time from the symptoms onset [3], [4] (shown in Fig. 1(a)). The accurate diagnosis of these stages essential in clinical practice to timely enable suitable life-saving treatment strategies (fibrinolytic therapy (FibT), percutaneous coronary intervention (PCI), and coronary artery bypass graft (CABG)), thereby improving viable myocardium salvage and prognosis [3], [4], [6]. However, existing approaches focus only on identifying MI patients from healthy controls (HC) using electrocardiogram (ECG) signals and do not address the diagnosis of MI severity stages. Also, these methods do not incorporate an additional non-MI class during the classification process to accommodate patients with CVDs other than MI. The inclusion of the non-MI class is essential to prevent misdiagnosis, thereby reducing over/under treatment [10]. To fill this gap, we propose an approach to classify the three MI severity stages from non-MI patients and HC subjects.

A. Clinical Background of MI Severity Stages

As shown in Fig. 1(a), reperfusion of the occluded artery within 40 minutes by FibT can salvage complete myocardium

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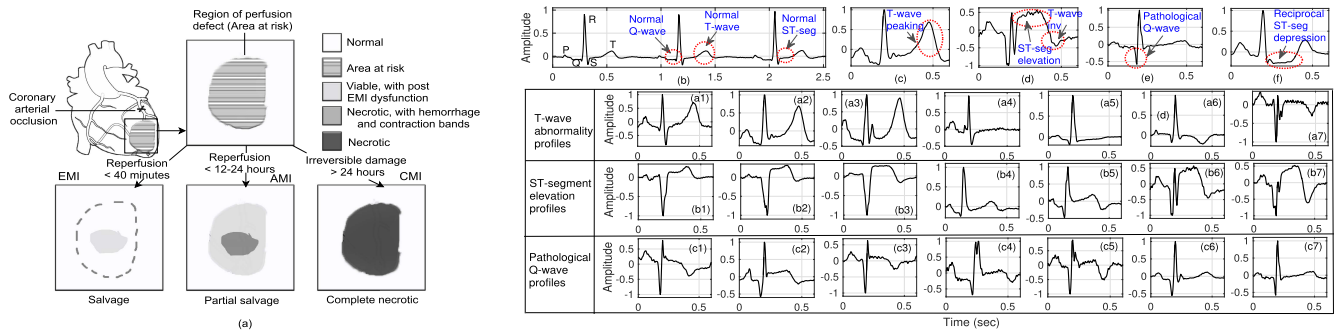


Fig. 1. (a) Illustrates the consequences of reperfusion at different MI severity stages such as EMI, AMI, and CMI [3], [4]. (b) Normal sinus rhythm, (c)-(f) myocardial infarction ECG characteristics. (a1)-(c7) Multi-scaled pathological ECG abnormality profiles of MI. (a1)-(a7) T-wave, (b1)-(b7) ST-elevation, and (c1)-(c7) pathological Q-wave abnormality profiles, respectively.

at risk [3]. This stage of MI is reversible and is called EMI [3], [7]. If the occlusion persists for 12-24 hours, cell death starts and causes irreversible myocardial damage. This stage of MI is called AMI and reperfusion with PCI or CABG, can help in partial myocardial salvage [3] (Fig. 1(a)). If occlusion still persists (> 24 hours) without treatment, the myocardium at risk becomes completely necrotic [3], [4]. This irreversible state of damage is known as CMI. It causes severe ventricular remodeling with more than 90% mortality [4].

According to the American Heart Association (AHA) guidelines, the 12-lead ECGs are preferred in clinical practice to diagnose MI [4], [8]. ECG is a non-invasive way of measuring surface potentials induced by cardiac electrical activity over time [2]. It is a simple and reliable tool for continuous monitoring of cardiac changes, either in-hospital or pre-hospital settings [8]. A typical ECG beat consists of P-wave, QRS-complex, and T-wave [2] shown in Fig. 1(b). During MI progression, these features undergo subtle changes like T-wave peaking and inversions, ST-elevations, and pathological Q-waves [2] (Fig. 1(b)-(e)). Specifically, the diagnosis of EMI is more sensitive to the T-wave abnormalities (peaking and inversions) shown in Fig. 1(c)-(d) with slight ST-elevations [14] (Fig. 1(b1)). In the AMI stage, as the necrosis starts, the beginning of pathological Q-waves, severe ST-elevations, and post-ischemic T-wave inversions appear in ECGs, as shown in Fig. 1(d), (b1)-(b2), and (b6)-(b7). Deep pathological Q-waves, normalizing ST-elevations and T-wave inversions [4], [6] appear in CMI stage (Fig. 1(e) and (c1)-(c7)). Generally, reciprocal ST-depressions (Fig. 1(f)) can also be seen in the leads opposite to the ST-elevation leads.

Manual examination of these subtle changes in ECG across the 12-leads is tedious, time-consuming and may suffer from intra-, and inter-observer variabilities [8], [9]. Therefore, research on the automated MI diagnosis systems has gained popularity as these systems can be easily deployed in various applications like in-hospital diagnosis and home-based personal healthcare for improved disease management.

B. Related Works

Most of the existing MI detection studies focus only on identifying MI patients from HC using ECGs [8]. These studies are broadly categorized into machine learning (ML), and deep learning (DL) approaches. The ML-based

frameworks consist of hand-crafted feature extraction, selection, and classification. In the feature extraction stage, time-domain [12]–[14] and transform domain [15]–[20] features are mostly employed. The commonly used classifiers are threshold-based [15], K-nearest neighbour (KNN) [12], [17], support vector machines (SVM) [17]–[19], random forest [13], and neural network (NN) [23]. Although these methods show good results on smaller datasets, they have some pertinent limitations. Firstly, feature extraction depends on the medical expertise, and the quantification of progressive MI changes using fixed hand-crafted features is difficult [24]. Secondly, the performance of the time-domain based methods depends on the robustness of fiducial points detection and often results in false MI alarms in the presence of various noises [26].

To overcome these limitations, the DL based models are employed to detect MI from ECGs. Unlike ML methods, the DL models do not require manual feature extraction as they automatically discover high-level feature representations needed for optimal diagnosis from the raw data [24]. Studies across various domains [24], [27], [40] show that DL models have better generalization capability compared to ML methods. The 1D-convolutional neural network (CNN) based DL models have recently been employed for MI detection using single-, limited- or 12-lead ECGs [28]–[34].

The *Single-lead models* use only one-lead ECG like Lead II [28] for detection. *Limited-lead models* target to identify MI at a specific portion of the heart by apriori selected leads from 12-lead ECG. In [29], [30] 3-leads have been used for the detection of inferior and anterior MIs. Generalized anterior MI detection using 4-leads has been reported in [31]. However, apriori selection of leads for specific localized MIs cannot be reliable as in real-world applications, patients can have MI at any location of the heart [4]. Also, these approaches do not adhere to the AHA guidelines of using 12-leads for MI diagnosis. Thus, these methods are not suitable to deploy in practical diagnostic systems. The *12-lead models* [32], [33] have exploited the 12-lead ECGs for MI diagnosis. In [32], a multiple-feature branch CNN (MFB-CNN) model has been proposed to comprehensively represent 12-lead ECGs for diagnosis. A hybrid network using CNN and bi-directional recurrent neural networks (BRNN) called MFB-CRNN been employed in [33]. Here, the MFB-CNN architecture is used for high-level feature extraction and BRNN used for feature

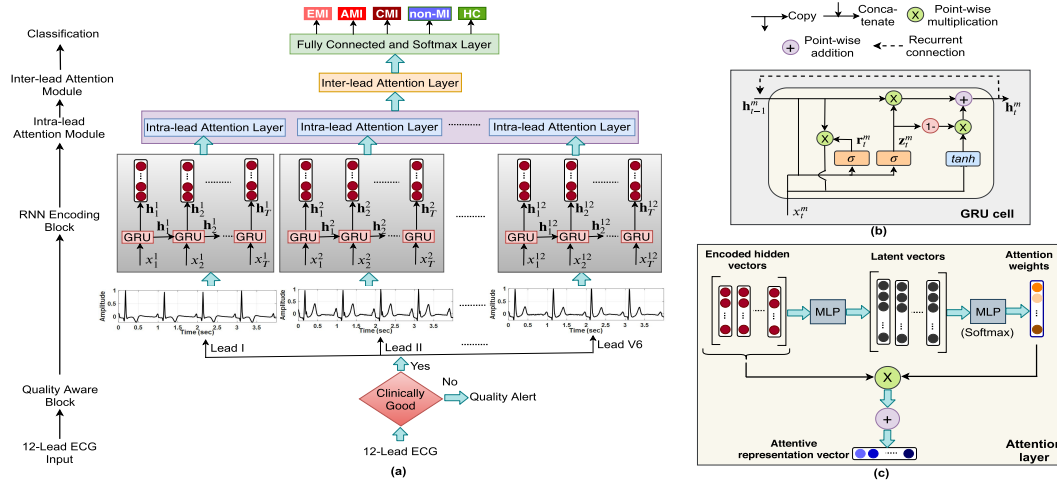


Fig. 2. (a) Overall view of proposed end-to-end classification model. (b) GRU cell. (c) self-attention layer.

summarization to improve diagnosis. Although these CNN based approaches show good performance for MI detection, they fail to capture the temporal dependencies of multi-scaled pathological ECG features present in the various MI severity stages (see Fig. 1(c)-(c7)). The existing methods also lack model interpretability, i.e., the features from the input ECG signals resulting in the classification decision are not known. This limits the applicability of the existing methods in clinical diagnostic systems [36].

Therefore, in this work, we explore the recurrent neural networks (RNN) to effectively model the sequential ECG data [38], [40]. Recently, the attention models have been employed in neural machine translation (NMT) [40] and image captioning [41] to discover informative regions to attend for optimal performance. Therefore, we adapted the attention mechanism for selecting clinically relevant within and across lead ECG information for improved diagnosis and interpretability. The main contributions of this paper are (1) design of a new multi-class diagnosis approach for identifying MI severity stages from non-MI disease patients and HCs to timely initiate suitable treatment strategies, (2) development of a unique end-to-end multi-lead diagnostic attention-based RNN (MLDA-RNN) model to effectively capture multi-scaled temporal dependencies and systematically process the 12-lead ECGs through intra- and inter-lead attention modules, and (3) analysis of attention weights for the model interpretability to find its correlation with clinicians while diagnosing various MI stages. The rest of the paper is organized as follows. Section II presents the proposed MLDA-RNN model for MI severity stages classification. Experimental results are provided in Section III. Finally, Section IV concludes the paper.

II. METHODOLOGY

Fig. 2(a) presents an overview of the proposed end-to-end MLDA-RNN model. It mainly consists of five processing blocks: the quality-aware block, the RNN encoding block, the intra-lead attention module, the inter-lead attention module, and the classification block. The detailed description of the blocks is given in the following Sections.

A. Quality Aware Block

In this stage, clinically good quality 12-lead ECGs are identified using the signal quality estimation method from [42] and allowed for further processing. It reduces frequent false MI alarms in the presence of various physiological and non-physiological noises [11], [26].

B. Recurrent Neural Network Encoding Block

RNNs are widely used for modeling sequential data. It contains recurrent connections, in which the activation of the current hidden state depends on the previous time step [27]. In this way, the network captures the dynamic temporal behavior well. However, it is difficult to train the vanilla RNNs for capturing long-term dependencies in data due to the exploding or vanishing gradient problem [27]. To alleviate this, long-short term memory (LSTM) and gated recurrent unit (GRU) structures are often used [37]. Both contain sophisticated activations and gating functions to regulate the flow of information to maintain proper gradients till backward steps [37]. Compared to LSTM, GRU has less trainable parameters with reduced computational complexity and is well suited for tasks which have limited datasets [37]. Thus, in this paper, we chose GRU as the basic RNN unit (see Fig. 2(b)).

Let $\mathbf{x}^m = [x_1^m, x_2^m, \dots, x_T^m]$ be the 12-lead ECG sequence data, where $m \in \{1, \dots, 12\}$ represents the lead number, and $x_t^m, t \in \{1, 2, \dots, T\}$ denotes m^{th} ECG lead information at the t^{th} time step and T is the sequence length. As shown in Fig. 2(b), GRU contains two important gates known as the reset gate (r_t^m) and the update gate (z_t^m) for controlling the flow of information inside the cell. r_t^m determines the part of the past information, which will be maintained in the current hidden state, z_t^m determines the part of new information, which will be added to the current hidden state. The activation of hidden layer for the given m^{th} ECG-lead at time step t is given by

$$\mathbf{h}_t^m = (1 - z_t^m) \odot \mathbf{h}_{t-1}^m + z_t^m \odot \tilde{\mathbf{h}}_t^m \quad (1)$$

where $\odot$ denotes element-wise multiplication. $\mathbf{z}_t^m$ is the update gate and it can be derived as

$$\mathbf{z}_t^m = \sigma(\mathbf{W}_{zi}^m x_t^m + \mathbf{W}_{zh}^m \mathbf{h}_{t-1}^m + \mathbf{b}_z^m) \quad (2)$$

where $\sigma(\cdot)$ is the sigmoid activation function, $\mathbf{b}_z^m$ is a bias vector, $\mathbf{W}_{zi}^m$ and $\mathbf{W}_{zh}^m$ represent the input-update and update-hidden weight matrices respectively. The candidate hidden state $\tilde{\mathbf{h}}_t^m$ can be computed as

$$\tilde{\mathbf{h}}_t^m = \tanh(\mathbf{W}_{ci}^m x_t^m + \mathbf{W}_{cr}^m (\mathbf{r}_t^m \odot \mathbf{h}_{t-1}^m) + \mathbf{b}_c^m) \quad (3)$$

where $\tanh$ is hyperbolic tangent activation function, $\odot$ denotes element-wise multiplication, $\mathbf{b}_c^m$ is bias vector, $\mathbf{W}_{ci}^m$ and $\mathbf{W}_{cr}^m$ denote the input-candidate, reset-candidate weight matrices. Finally, the reset gate vector $\mathbf{r}_t^m$ can be given by

$$\mathbf{r}_t^m = \sigma(\mathbf{W}_{ri}^m x_t^m + \mathbf{W}_{rh}^m \mathbf{h}_{t-1}^m + \mathbf{b}_r^m) \quad (4)$$

where $\sigma(\cdot)$ is a sigmoid activation, $\mathbf{b}_r^m$ is bias vector, $\mathbf{W}_{ri}^m$ and $\mathbf{W}_{rh}^m$ denotes the input-reset, reset-hidden weight matrices.

Finally, for m^{th} ECG-lead $x_t^m, t \in \{1, 2, \dots, T\}$ its GRU encoded hidden vectors are given as $\mathbf{h}_t^m, t \in \{1, 2, \dots, T\}$. $\mathbf{H}_m \in \mathbb{R}^{d_h \times T}$ contains the m^{th} -lead ECG hidden vectors across all time steps obtained from RNN encoding block as

$$\mathbf{H}_m = [\mathbf{h}_1^m, \mathbf{h}_2^m, \dots, \mathbf{h}_t^m, \dots, \mathbf{h}_T^m] \quad (5)$$

where $\mathbf{h}_t^m \in \mathbb{R}^{d_h}$ represents m^{th} ECG lead hidden vector at current time step t , d_h denotes size of the hidden vector reflect GRU memory, and T is sequence length.

Since all the 12-lead ECGs are simultaneously acquired from the same cardiac source, they show implicit intra- and inter-lead redundancies [2]. This often results in feature overlapping and limits the classification performance [8]. Although GRU improves learning for such data, it encodes the entire sequential information into a fixed-size vector before it is fed to the classifier. This representation may be too abbreviated to capture subtle changes of ECG for progressive diseases like MI [39]. Attention mechanism has shown promising results to handle such a problem [40], [41]. Thus, the obtained hidden vectors of each ECG lead from the RNN encoding blocks are fed to the respective intra-lead attention layers.

C. Intra-Lead Attention Module

This module discovers the clinically relevant hidden vectors corresponding to the ECG features essential to improve the overall classification. Fig. 2(c) shows the network architecture of the attention layer present in this module. Each encoded hidden vectors $\mathbf{h}_t^m$ obtained from the RNN encoding block are first non-linearly transformed into a latent space using a multi-layer perceptron (MLP);

$$\mathbf{u}_t^m = \tanh(\mathbf{W}_{intra}^m \mathbf{h}_t^m + \mathbf{b}_{intra}^m) \quad (6)$$

where $\mathbf{W}_{intra}^m \in \mathbb{R}^{d_a \times d_h}$ and $\mathbf{b}_{intra}^m \in \mathbb{R}^{d_a}$ are the weight matrix and the bias vector of a MLP with latent size d_a , to obtain $\mathbf{u}_t^m \in \mathbb{R}^{d_a}$ as latent representation of $\mathbf{h}_t^m$. These latent vectors ($\mathbf{u}_t^m$) are fed to a MLP (dimension one) with Softmax

activation, to obtain the weights of importance for each hidden vector $\mathbf{h}_t^m$ as

$$\alpha_t^m = \frac{\exp((\mathbf{u}_t^m)^T \mathbf{u}_{intra}^m)}{\sum_{t=1}^T \exp((\mathbf{u}_t^m)^T \mathbf{u}_{intra}^m)} \quad (7)$$

where $\mathbf{u}_{intra}^m \in \mathbb{R}^{d_a}$ corresponds to the intra-lead hidden attention vector. The dot-product similarity between every time instants $\mathbf{u}_t^m$ and $\mathbf{u}_{intra}^m$ provides intra-lead attention weight α_t^m at each instant corresponding to the m^{th} ECG lead. We jointly learn $\mathbf{W}_{intra}^m$, $\mathbf{b}_{intra}^m$, and $\mathbf{u}_{intra}^m$ parameters along with the other model parameters during training process. The *Softmax*($\cdot$) function guarantees that all the computed weights sum to 1, and these weights will be relatively high for the hidden vectors, which are clinically more relevant for classification than others. Lastly, the weighted sum of all the temporal hidden vectors of each lead is calculated to form a high-level representation called within-lead attentive representation (WLAR) vector given as

$$\mathbf{a}_m = \sum_{t=1}^T \alpha_t^m \mathbf{h}_t^m \quad (8)$$

where $\mathbf{a}_m \in \mathbb{R}^{d_h}$ denotes the WLAR vector for m^{th} ECG lead. Twelve such vectors (from 12-leads) are used to construct the within-lead attentive matrix $\mathbf{A} \in \mathbb{R}^{d_h \times 12}$ and is fed to the inter-lead attention module.

D. Inter-Lead Attention Module

Clinically, MI diagnosis using 12-lead ECG involves observing intra-lead diagnostic features followed by summarization across all 12 leads to make a final decision [4]. Therefore, the inter-lead attention module effectively aggregates the obtained WLAR vectors corresponding to the 12-lead ECGs based on their importance in the overall classification. This summarization gives a high-level across-lead attentive representation (ALAR) vector denoted as $\mathbf{a}_f$ and is computed with a similar formulation as discussed in Section II.C;

$$\mathbf{v}_m = \tanh(\mathbf{W}_{inter} \mathbf{a}_m + \mathbf{b}_{inter}) \quad (9)$$

$$\beta_m = \frac{\exp((\mathbf{v}_m)^T \mathbf{v}_{inter})}{\sum_{m=1}^{12} \exp((\mathbf{v}_m)^T \mathbf{v}_{inter})} \quad (10)$$

$$\mathbf{a}_f = \sum_{m=1}^{12} \beta_m \mathbf{a}_m \quad (11)$$

where $\mathbf{a}_f \in \mathbb{R}^{d_h}$ is the final ALAR vector with emphasis given to specific ECG leads relevant for classification thereby reducing inter-lead redundancy. $\mathbf{W}_{inter}$, $\mathbf{b}_{inter}$ are weight matrix and bias vector of a MLP whose latent size is d_a that transforms $\mathbf{a}_m \in \mathbb{R}^{d_h}$ to $\mathbf{v}_m \in \mathbb{R}^{d_a}$. The dot-product between $\mathbf{v}_m$ and the inter-lead attention vector $\mathbf{v}_{inter}$ gives weight of importance β_m for each ECG lead. We jointly learn $\mathbf{W}_{inter}$, $\mathbf{b}_{inter}$, and $\mathbf{v}_{inter}$ parameters along with the other model parameters during training process. The compact representation vector $\mathbf{a}_f$ is next fed to the classification block.

E. Classification

Lastly, the $\mathbf{a}_f$ is fed to fully-connected layer followed by softmax activation (FC+Softmax), to estimate probability distribution over different classes denoted by $\hat{\mathbf{y}}$ as

$$\hat{\mathbf{y}} = \text{Softmax}(\mathbf{W}_o \mathbf{a}_f + \mathbf{b}_o) \quad (12)$$

where $\mathbf{W}_o \in \mathbb{R}^{K \times d_h}$ and $\mathbf{b}_o \in \mathbb{R}^K$ are weight matrix and bias vector of an output layer and K is the number classes under study. The predicted class label from $\hat{\mathbf{y}} \in \mathbb{R}^K$ can be obtained by $\text{argmax}(\hat{\mathbf{y}})$.

All the parameters in the network are trained using multi-class categorical cross-entropy loss ($\mathcal{L}$) over all training samples and classes [27] computed as

$$\mathcal{L} = -\frac{1}{N} \sum_{n=1}^N \sum_{k=1}^K (\mathbf{y}_{n,k}) \log(\hat{\mathbf{y}}_{n,k}) \quad (13)$$

where N is the number of training samples, and $\mathbf{y}_{nk}$ and $\hat{\mathbf{y}}_{nk}$ are true and predicted labels of n^{th} training sample. In the next Section, we provide details of the network architecture and experimental results.

III. EXPERIMENTAL RESULTS AND DISCUSSION

In this Section, we discuss the details of the dataset used and the evaluation scheme. We also present the details of the network optimization followed by the experimental evaluation and analysis of the proposed MLDA-RNN model.

A. Data Set

The proposed method is evaluated on the 12-lead ECG signals obtained from the publicly available STAFF III and PTB diagnostic databases [43]. In this work, the input ECG length is chosen to be 4 sec as progressive diseases like MI require an ample number of consecutive beats for analysis [18]. It allows us to capture the intra-, and inter-beat dependencies well. The specific details of the databases are given below.

1) **STAFF III Database:** The ECG data for EMI patients are selected from this database. STAFF III provides induced early progression of MI (EMI) data from patients receiving elective percutaneous transluminal angiography in one of the major coronary arteries. The database consists of 108 patients with nine leads (V1-V6, I, II, and III) at a sampling rate (f_s) of 1000 Hz. The remaining three leads (aVR, aVL, and aVF) are computed from limb leads to form standard 12-lead ECG.

2) **PTB Diagnostic Database:** The ECG data for the AMI, CMI, non-MI, and HCs are obtained from this database. It consists of 549 records from 290 individuals with 148 MI patients, 52 HCs, and the rest are non-MI. Each record contains 12-lead ECGs with f_s 1000 Hz. From the patient annotations, based on the infarct age and interventions, we selected data for AMI and CMI patients with infarct age less than 12-24 hours, and more than 24 hours, respectively [3].

To obtain well representative MI data, we excluded prior MI patients from both the databases. All the ECG signals are downsampled to 125 Hz. Table I shows the selected dataset.

TABLE I
STATISTICS OF THE ECG DATA USED

Class	Database	No.of subjects	Selected sub.	No.of male/female	No.of 12-lead 4 sec segments
EMI	STAFF III	108	52	38/14	3609
AMI	PTB	148	55	42/13	2107
CMI	PTB	148	109	83/26	3618
non-MI	PTB	64	64	40/24	1902
HC	PTB	52	52	39/13	3024
Total	-	372	332	242/90	14260

B. Evaluation Scheme and Performance Measures

Class-oriented and patient-independent are two evaluation schemes often used in ECG based disease classification tasks [32]. In the class-oriented scheme, ECG signals from training and test sets may belong to the same patient, which often results in poor generalization for signals from the unseen patients [25]. On the other hand, a patient-independent evaluation scheme [31] ensures no overlap of patient data while dividing train and test datasets. Therefore, we employ a patient-independent scheme for evaluation. The classification performance of the proposed method is evaluated by the overall sensitivity (OS), specificity (OSp), and accuracy (OA) along with class-wise sensitivity (Se), specificity (Sp), and accuracy (Acc) [8]. 5-fold cross-validation (CV) is employed to avoid overfitting for a specified dataset [27].

C. Network Architecture

Each lead ECG with 500 samples ($T = 500$), are encoded with single layer RNN of size 32 ($d_h = 32$) to obtain hidden vectors. These vectors are fed to the respective intra-lead attention module with latent space dimension 64 ($d_a = 64$) to form WLAR vectors ($\mathbf{a}_m$). These WLAR vectors obtained by emphasizing within lead diagnostic information, thereby reducing intra-lead redundancies. Later, these WLAR vectors are fed to inter-lead attention module with latent space size 64 ($d_a = 64$) to generate ALAR vector ($\mathbf{a}_f$). This vector attends the most informative leads, thereby reducing inter-lead redundancy. This reduction of intra- and inter-lead redundancies will improve the classification by generating most discriminative high-level representations. Finally, the 32-dimensional ALAR vector is fed to the FC+Softmax layer to classify the MI severity stages. To improve the speed of convergence and performance of the model [27], we employed batch normalization (BN) layers for each lead after the inter-lead attention module, and after the inter-lead attention module. A dropout layer, with a drop rate of 0.3 is used before the output layer to improve the model generalization [37].

The hyperparameters, RNN encoding size (d_h) and latent-space size (d_a) used in the intra- and inter-lead attention module are optimized using grid search. To test the effect of these parameters on the classification, we fix the number of RNN layers to one and select d_h from the set $\{4, 8, \dots, 128\}$. Then we fix d_h and choose d_a from the set $\{4, 8, \dots, 256\}$. From Fig. 3, it can be observed that the proposed method achieves better OA when $d_h \geq 32$ and $d_a \geq 64$ than the

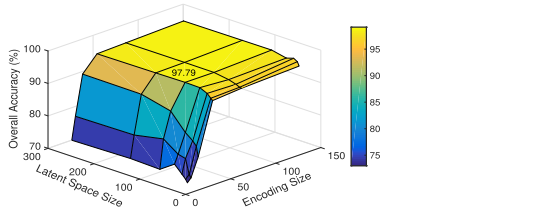


Fig. 3. Performance variation of the MLDA-RNN model with different encoding size and latent space size.

TABLE II

THE AVERAGE CLASS-WISE AND OVERALL PERFORMANCE MEASURES WITH THE STANDARD DEVIATION VALUES (5 FOLDS) FOR THE PROPOSED METHOD FOR THE ENTIRE DATASET

Class	Se (%)	Sp (%)	Acc (%)	OS/OSp/OA (%)
EMI	98.34 ± 0.10	99.43 ± 0.08	99.15 ± 0.26	97.60 ± 0.54
AMI	94.78 ± 0.88	99.62 ± 0.11	98.91 ± 0.29	
CMI	97.64 ± 0.34	98.87 ± 0.06	98.56 ± 0.33	
non-MI	97.90 ± 0.29	99.71 ± 0.06	99.47 ± 0.02	
HC	99.33 ± 0.38	99.51 ± 0.14	99.47 ± 0.26	97.79 ± 0.47

other values. The OA is poor at lower values of d_h due to not having sufficient memory. Once enough memory (≥ 32) has been employed, OA increases to 97.79 % at d_a equal to 64. It can also be seen that, when we increase d_h and d_a beyond 64 and 128, due to an increase in complexity of the network for the available dataset, the network starts over-fitting [33]. Thus, from this analysis, the optimized d_h and d_a values are 32 and 64 respectively.

The whole network is developed using Keras with a TensorFlow backend. The multi-class loss is calculated using (13) and the model is trained using backpropagation through time (BPTT) [27]. The optimization method employed is an adaptive moment estimation algorithm with a learning rate of 0.001. Batch size is set to 32 with a number of epochs 200 along with early stopping criteria to reduce over-fitting of the model for unseen patients [37].

D. Evaluation of the Proposed Method

Table II shows the class-wise and overall performance measures for the proposed method. It can be observed that the proposed method attains an impressive OS and OA of 97.60% and 97.79%, respectively. The method also shows promising results for confusing classes like the AMI and CMI with Se of 94.78% and 97.64%, respectively. This can also be verified by the reduced false negatives as shown in the confusion matrix (Table III). The slight reduction of Se (Table II) for these classes compared to EMI and HC can be attributed to the subtle differences in the ECG characteristics and the inter-patient variability [30]. Similarly, class-wise Sp and Acc for all classes are well above 98% indicates that the proposed model is near accurate in identifying true negatives (Table III). It can also be observed from the Table II that with small variance in measures across 5-folds for all classes shows an excellent generalization of the proposed MLDA-RNN model. Finally, Tables II and III reflect the improved overall performance of the proposed model without compromising on the

TABLE III

CONFUSION MATRIX OF PROPOSED METHOD ACROSS 5 FOLDS

Actual/Predicted	EMI	AMI	CMI	non-MI	HC
EMI	3544	20	26	5	15
AMI	9	2001	78	22	0
CMI	11	24	3528	10	42
non-MI	33	0	12	1860	0
HC	11	0	10	1	2998

TABLE IV

THE AVERAGE PERFORMANCE COMPARISON OF THE RNN-I, RNN-II, AND PROPOSED MODELS FOR THE ENTIRE DATASET (5 FOLDS)

Method	Class	Se (%)	Sp (%)	Acc (%)	OS/OSp/OA (%)
RNN-I (no attention)	EMI	94.04	94.92	94.70	71.67 93.80 76.26
	AMI	94.7	98.97	85.72	
	CMI	82.84	82.94	82.92	
	non-MI	81.62	97.77	95.61	
	HC	90.39	94.38	93.54	
RNN-II (only intra-lead attention)	EMI	95.42	99.20	98.24	90.52 97.74 91.08
	AMI	77.01	96.29	93.43	
	CMI	87.41	95.01	93.08	
	non-MI	95.27	99.43	98.87	
	HC	97.50	98.79	98.52	
Proposed (intra- and inter-lead attention)	EMI	98.34	99.43	99.15	97.60 99.43 97.79
	AMI	94.78	99.62	98.91	
	CMI	97.64	98.87	98.56	
	non-MI	97.90	99.71	99.47	
	HC	99.33	99.51	99.47	

class-wise detection rates. This can be attributed to the RNNs with intra- and inter-lead attention modules to effectively capture the subtle discriminative ECG features for improved diagnosis.

E. Effectiveness of the Proposed MLDA-RNN Model

In this Section, we analyze the effectiveness of the proposed MLDA-RNN model from two perspectives; (i) significance of the intra-, and inter-lead attention modules and (ii) model interpretability using intra-, and inter-lead attention weights.

1) *Significance of Intra- and Inter-Lead Attention Modules:* Table IV summarizes the performance comparison of the proposed model having intra- and inter-lead attention modules with models having no attention (RNN-I) and with only intra-lead attention (RNN-II). These two models follow the same architecture as the proposed model (Fig. 2(a)) with a slight difference in the way the high-level representation vector generated before giving it the classifier. The RNN-I uses the last hidden vector from each lead RNN encoding block and averages them across 12-leads before feeding it to the classification stage. On the other hand, the RNN-II with intra-lead attention module combines within-lead relevant hidden vectors to form WLAR, and these vectors are averaged across all the 12-leads and fed to the classification stage.

From Table IV, it can be observed that the proposed model outperforms RNN-I and RNN-II models in terms of various measures. The RNN-I model shows poor Se for CMI, non-MI, HC classes, and very poor for the AMI class. However, the intra-lead attention module in RNN-II helps to improve the performance for all classes considerably. Still, some patients from the AMI and CMI classes are misdiagnosed. However, the proposed method helps to improve performance for all

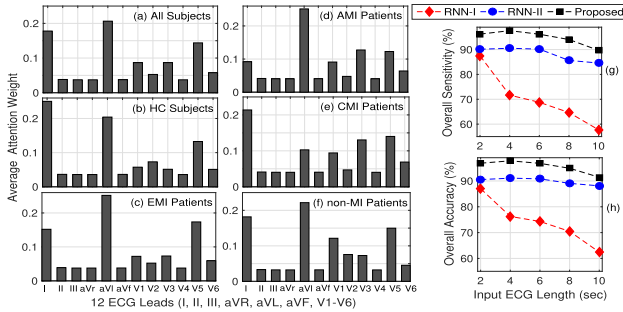


Fig. 4. (a)-(f) Lead-wise average attention weights for all subjects, HC, EMI, AMI, CMI and non-MI patients, respectively. (g) and (h) overall sensitivity and accuracy comparison for RNN-I, RNN-II, and proposed model over different input ECG lengths.

classes concerning class-wise and overall measures. As shown in Table IV, from RNN-I to the proposed model, there is a significant improvement for the overall measures OS , OSp , and OA of about 26%, 5%, and 21%, respectively. Therefore, intra- and inter-lead attention modules in the proposed model contribute individually to improve the performance by attending relevant within and across leads information, thereby reducing intra- and inter-lead redundancies.

We also performed experiments on different ECG input lengths to show the effectiveness of the proposed method for long-term ECG signals. Fig. 4(g)-(h) shows OS and OA measures for RNN-I, RNN-II, and the proposed method across different ECG input lengths. It can be seen that the proposed model outperforms RNN-I and RNN-II irrespective of ECG length. Even for long-duration ECGs of 10 sec, the OS and OA are reasonably good as nearly 90% and 91%, respectively. This shows the effectiveness of the proposed model for analyzing progressive diseases having long-term dependencies in ECG. It can also be observed that the highest OA for the proposed method is obtained at an input length of 4 seconds. This verifies the choice of selecting 4 seconds duration ECG signals for the proposed framework.

2) Model Interpretability Using Intra- and Inter-Lead Attention Weights: Model explainability is as important as accuracy in the healthcare applications, which most of the existing DL based methods lack [27], [36]. The certainty needs to exist that the diagnosis of the disease is made based on real ECG characteristics that reflect pathology rather than systematic biases in the dataset. From this perspective, the proposed model's intra- and inter-lead attention weights are visualized and analyzed for different classes in the present Section.

Fig. 4(a)-(f) show the lead-wise average attention weights for all classes along with all subjects of the dataset. It can be observed that out of six frontal plane ECG leads, the model learns to give more weight to the leads I and aVL. The physiological reason for giving more weight to these leads includes the fact that they monitor the left side of the heart and are critical for the diagnosis of MI and non-MI patients [2], [4]. Among the six chest leads, V1, V3, V5, and V6 are given more importance while combining 12-lead ECG information as these leads are physiologically crucial in identifying various localized MIs (for example, V1 is sensitive for septal MI, V3 for anterior wall MI, V5 and V6 are for lateral wall MI) [4], [31]. For non-MI patients, more emphasis is given to the

same leads as these are very critical for diagnosing ventricular hypertrophy and bundle branch block (BBB) [4]. In summary, the proposed model attend specific leads while combining 12-lead ECG information essential for the improved classification. This often correlates with the clinicians' way of looking into informative leads for the diagnosis [4], [31].

Fig. 5 shows the mapping of intra-lead attention weights on to the most attended ECG leads (I, aVL, V3, and V5) for the different classes. It is observed that more attention is given to the ECG features that are sensitive to MI and non-MI pathologies. From Fig. 5(b1)-(b4) for EMI patient (with septal MI), it can be seen that ST-segment in the lead I, QRS-ST segment in aVL, J and J+80msec points in V5, and J-point with a portion of ST-segments in V6 are given more importance [8]. Similar attention can also be seen in Fig. 5(a1)-(a4), (c1)-(c4) for HC subject and AMI patient (with anterior-septal MI) respectively. Particularly, for CMI patient (with anterior-septal MI) shown in Fig. 5(d1)-(d4), pathological Q-waves has been given more importance along with normalized ST and T-wave variations [8]. For non-MI class (LBBB patient) shown in Fig. 5(e1)-(e4), it is observed that the broad and deep S-waves in leads I, V2-V3, and V5 along with broad R-waves in V5 are given more importance during classification [4]. From this analysis, it is evident that the proposed model is often attending ECG features that are clinically relevant for identifying various MI severity stages from non-MI patients and HCs [2], [4]. This often agrees with the clinicians' way of examining specific ECG characteristics for the diagnosis.

F. Comparison With the State-of-the-Art Methods

In this Section we compare the proposed MLDA-RNN model with the existing methods on various aspects. As can be seen from Table V, all the existing methods have performed the identification of MI from HCs only. However, we addressed the classification of MI severity stages, which is essential to optimize various treatment strategies. Secondly, some of the existing methods [28]–[31] use limited lead ECGs for MI or specific localized MI detection. It can be seen from Table V that these methods result in poor generalization, and it is evident from [29], [30] as average accuracy is below 85%. As discussed earlier, these approaches may not provide a reliable diagnosis for various localized MIs [4]. Table V also highlights the dependency of the existing methods on R-peak detection. It can be seen that most of the methods require a separate R-peak detection as a pre-processing step. This increases the computational burden and may also be prone to erroneous diagnosis in the presence of various noises [26].

The comparison methods in Table V can be broadly categorized into the ML and DL based methods. The pertinent drawbacks of the ML-based methods [18], [19], [24], have already been discussed in the previous Sections. The DL based approaches [28]–[34] mitigate these challenges and provide better generalization performance. Therefore, we implemented some recent DL based models for the proposed multi-class task to perform a quantitative comparison. These models are; (i) CNN based framework using lead V4 ECG signals (1L-CNN) reported by

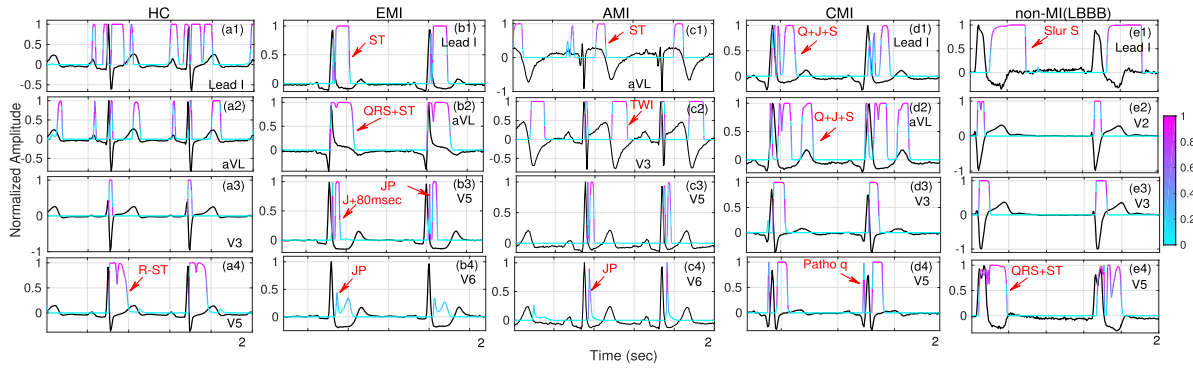


Fig. 5. Mapping intra-lead attention weights of hidden vectors to ECG leads for different classes (first 2 sec), (a1)-(a4) for HC subject, (b1)-(b4) for EMI, (c1)-(c4) for AMI, (d1)-(d4) for CMI and (e1)-(e4) for LBBB patients, respectively. Colour bar indicates the intensity magnitude proportional to clinical relevance (JP: J-point; J+80msec: 80 msec after J-point; ST: ST-segment; Q+J+S: QRS, J-point and ST segment; QRS+ST: QRS and ST-segment; R-ST: downsloping R-S segment and ST-segment; TWI: T-wave inversion; Patho q: Pathological Q-wave; Slur S: slurred S-wave).

TABLE V
COMPARISON OF THE PROPOSED METHOD AGAINST EXISTING METHODS ON MI DETECTION USING PTB DATASET*

Method (year)	#classes	# leads	R-peak detection	Decision	Selected dataset	#features	Classifier	Results	Evaluation	Risk stratification
[18] (2015)	2 (MI, HC)	12	yes	segment	MI:100 HC:100	72(multiscale energy and eigen value features)	SVM	Se=93% Sp=99.00% Acc=96.00%	class based	×
[19] (2019)	2 (MI, HC)	12	no	segment	MI:100 HC:74	108(empirical WT)	SVM	Se=100% Sp=99.95% Acc=99.97%	class based	×
[22] (2019)	2 (MI, HC)	12	yes	beat	MI:28213 HC:5373	22(wavelet packet transform based)	SVM	Se=80.96% Sp=80.96% Acc=92.69%	class based	×
[28] (2017)	2 (MI, HC)	1	yes	beat	MI:40182 HC:10546	End-to-end CNN (with R-peak detection)	softmax	Se=93.71% Sp=92.83% Acc=93.53%	class based	×
[29] (2017)	2 (IMI, HC)	3	no	segment	IMI:3222 HC:3055	End-to-end CNN	softmax	Se=85.33% Sp=84.04% Acc=84.54%	patient-independent	×
[30] (2018)	2 (AMI, HC)	3	yes	beat	AMI:41087 HC:18640	End-to-end CNN (with R-peak detection)	softmax	Se=85.33% Sp=84.09% Acc=84.54%	class based	×
[31] (2018)	2 (GAMI, HC)	4	yes	beat	AMI:34769 HC:34769	fuzzy information granulation, CNN (with R-peak detection)	softmax	Se=95.40% Sp=97.37% Acc=96.00%	patient-independent	×
[32] (2018)	2 (MI, HC)	12	yes	beat	MI:48690 HC:10646	End-to-end MFB-CNN (with R-peak detection)	softmax	Se=99.97% Sp=99.90% Acc=99.95%	class based	×
[33] (2019)	2 (MI, HC)	12	yes	beat	MI:53712 HC:10638	End-to-end MFB-CBRNN (with R-peak detection)	softmax	Se=94.42% Sp=86.29% Acc=93.08%	patient-independent	×
[34] (2019)	2 (MI, HC)	12	yes	beat	MI:485752 HC:125652	End-to-end CNN (with R-peak detection)	softmax	Acc=99.78%	class based	×
Proposed	5 (EMI, AMI, CMI, non-MI, HC)	12	no	segment	EMI:3609 AMI:2107 CMI:3618 non-MI:1902 HC:3024	End-to-end intra- and inter-lead attention based RNN model	softmax	OS=97.60% OSp=99.43% OA=97.79% AUC=0.98 F1=97.65% kappa=0.97	patient-independent	✓

*: results from the published articles, #: number of; selected dataset numbers corresponds to either beats/segments according to type of decision

Acharya *et al.* [28], (ii) CNN model using 3-leads (II, III, aVF) developed by Reasat and Shahnaz [29] named 3L-CNN, and (iii) 12-lead ECG based models i.e., MFB-CNN [32] and MFB-CBRNN [33] frameworks developed by Liu *et al.* These models were initially proposed for the detection of MI from HC subjects. However, they have been optimized for our dataset with five classes.

The results of the quantitative comparison are shown in terms of Se , Sp , Acc , OS , OSp , and OA . We also present the area under the ROC curve (AUC), average F1-score ($F1$), and kappa values along with the average test run

time (ms) in Table VI for the proposed and the existing methods. Although 1L-CNN [28] and 3L-CNN [29] perform well, physiologically, limited-leads are not enough to diagnose various localized MIs reliably [4]. This is reflected in their poor class-wise and overall performance seen from Table VI. The MFB-CNN [32] and MFB-CBRNN [33] exploit 12-lead ECGs for improved diagnosis. It is observed from the results column of Table V that these models show promising performance for MI vs. HC classification. However, these CNN based models show degraded OA of 83.13% and 90.98% respectively for the proposed multi-class approach having subtle changes

TABLE VI
THE AVERAGE PERFORMANCE COMPARISON OF THE PROPOSED AND THE EXISTING DL METHODS FOR THE ENTIRE DATASET

Method	Class	Se (%)	Sp (%)	Acc (%)	OS/OSp/OA (%)	AUC/F1(%) /kappa	Run-time (ms)
1L-CNN [28]	EMI	99.03	99.62	99.47			
	AMI	77.96	98.02	99.47	91.78	0.95	
	CMI	87.69	94.69	92.91	98.01	92.25	0.71
	non-MI	98.68	99.71	99.57	92.25	0.90	
	HC	95.52	97.99	97.47			
3L-CNN [29]	EMI	98.61	98.82	98.77			
	AMI	86.25	81.64	82.32	77.07	0.84	
	CMI	41.63	96.38	82.50	93.98	75.14	2.80
	non-MI	86.87	97.77	96.31	75.14	0.68	
	HC	72.01	95.28	90.35			
MFB-CNN [32]	EMI	92.24	96.29	95.26			
	AMI	62.32	93.37	88.77	82.25	0.89	
	CMI	76.07	92.86	88.60	95.72	83.13	1.58
	non-MI	89.23	99.51	98.14	83.13	0.78	
	HC	91.39	96.57	95.47			
MFB-CBRNN [33]	EMI	95.84	97.65	97.19			
	AMI	70.85	98.72	94.60	90.01	0.94	
	CMI	89.90	95.44	94.03	97.66	90.98	6.32
	non-MI	96.06	99.63	99.15	90.98	0.88	
	HC	97.35	96.88	96.98			
Proposed	EMI	98.34	99.43	99.15			
	AMI	94.78	99.62	98.91	97.60	0.98	
	CMI	97.64	98.87	98.56	99.43	97.65	8.76
	non-MI	97.90	99.71	99.47	97.79	0.97	
	HC	99.33	99.51	99.47			

in multi-scaled pathological ECG features (see Table VI). Compared to the recent state-of-the-art MFB-CBRNN model, the proposed MLDA-RNN shows a significant improvement of nearly 7% in *OA* without compromising on class-wise detection rates. The model also exhibits improved *AUC*, *F1* and kappa values of 0.98, 97.65%, and 0.97 respectively. It can also be seen from Table VI that the proposed method takes an average run time of 8.76 ms to generate a classification result (results taken from system with an i5 processor with 16 GB RAM and Quadro K600 graphics processor). This is slightly higher compared to the existing methods. However, clinically, 8.76 ms run time is not significantly high to delay the emergent interventions [9].

From the comparison results in Table V, it can be observed that the MLDA-RNN has impressive advantages over other algorithms. (i) *Clinically relevant problem*: we have addressed a novel MI severity stages classification problem where the discriminative changes in the ECG features are subtle. The automatic detection of these classes will help in guiding proper treatment plans, thereby reducing mortality. (ii) *Inclusion of non-MI class*: we considered other disease patients through the non-MI class to reduce over/under treatment. (iii) *DL Model*: the proposed MLDA-RNN exploited RNNs with intra-, and inter-lead attention modules, to capture the temporal dependencies and fuse diagnostically relevant within and across lead information for improved diagnosis. (iv) *Model interpretability*: MLDA-RNN shows impressive interpretability results which lack in existing DL models. It is visually and experimentally observed that the intra-, and inter-lead attention weights are often correlated with the clinicians' way of diagnosis. Considering these advantages, the proposed MLDA-RNN approach can reliably be used in pre-, and in-hospital settings for diagnosing MI severity stages and assisting cardiologists for initiating suitable emergent interventions.

IV. CONCLUSION

In this paper, we addressed a novel approach of identifying three MI severity stages from non-MI patients and HC subjects using 12-lead ECGs. For this task, we proposed an MLDA-RNN model with a unique architecture employing RNNs, intra- and inter-lead attention modules. These components can systematically process 12-lead ECGs to learn discriminative features by focusing on clinically relevant information for the improved classification. With improved performance, MLDA-RNN also shows promising results for the model interpretability as obtained attention weights often correlate with the cardiologists while diagnosing MI. The proposed approach addressed only the classification of MI severity stages. In the future, we will extend the proposed model to diagnose various localized MIs along with its severity. This can also be extended to the limited ECG data diseases in a transfer learning framework to make use of its knowledge of attending clinical features.

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